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The Oxalate Dianion, $C_2O_4^{2-}$: Planar or Nonplanar?

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ABSTRACT: Theoretical studies have shown that for the isolated oxalate dianion, $C_2O_4^{2-}$, which contains a long C−C single bond, the conformer with a 90° twist about the C−C bond (D_{2d} symmetry) is more stable than the planar (D_{2h}) conformer that is almost invariably drawn for the dianion. The D_{2d} conformer has been shown to persist in aqueous solutions of simple oxalate salts. However, in oxalato complexes or in solid oxalates, the O−C−C−O dihedral angle for $C_2O_4^{2-}$ is commonly observed to be quite different from the 90° of the D_{2d} conformer. This indicates that bonding or "crystal packing" forces usually outweigh the 2−6 kcal/mol barrier calculated for rotation between the two conformers.

KEYWORDS: *Second-Year Undergraduate, Upper-Division Undergraduate, Inorganic Chemistry, Interdisciplinary/Multidisciplinary, Misconceptions/Discrepant Events, Coordination Compounds, Descriptive Chemistry, Molecular Properties/Structure, Stereochemistry*

Many undergraduate students might be asked to draw a Lewis structure for the oxalate dianion, $C_2O_4^{2-}$ (ox^{2-}), which will, of necessity, be two-dimensional. Also, it is not uncommon for students to prepare an oxalato complex in an inorganic chemistry laboratory. The syntheses of several salts of such complexes are given in this *Journal*: $K_3[M(ox)_3]\cdot 3H_2O$, where M is Al[1] or Fe,[2] and $K_2[Cu(ox)_2]\cdot 2H_2O$.[3] Another salt that is commonly prepared by students is $K_3[Cr(ox)_3]\cdot 3H_2O$.[4,5] The student syntheses probably lead to their first (possibly only) exposure to a representation of the three-dimensional structure of the oxalate dianion. In the above salts X-ray crystallographic studies have shown that, as usual, the chelating oxalato ligands are planar or close to planar.[6−8] Similarly, bridging oxalato ligands are usually found to be planar or close to planar also, although examples of such ligands distinctly twisted about the C-to-C linkage (O−C−C−O dihedral twist angles 76° and 78°) do exist.[9] Thus, in diagrammatic representations, chelating (or bridging) ox^{2-} is usually shown as planar. Students may also learn that crystalline $H_2ox\cdot 2H_2O$, a common reagent in the syntheses of oxalato complexes, contains planar molecules of the acid.[10] Therefore it is perhaps not surprising that the oxalate dianion is usually represented as planar; for example it is shown in this way in a popular online encyclopedia.[11]

It is now generally agreed that in ox^{2-} a long C−C single bond occurs, that is, there is no π-bonding between the two planar CO_2 moieties, consistent with the ground-state Lewis structure and with MO treatments of the dianion in its planar form.[12−14] Thus, free rotation about the C−C bond of oxalate should be possible. In fact, several theoretical studies have shown that the most stable conformation of an isolated ox^{2-} is *not* planar (D_{2h} symmetry), but rather that with a 90° twist about the C−C bond (D_{2d} symmetry).[15−17] The atomic skeletons of the two extreme conformers are shown in Figure 1. Clark and Schleyer[15] attribute the stability of the D_{2d} conformer to predominant electrostatic effects. The planar conformer may be obtained from the more stable D_{2d} conformer by rotation about the C−C bond; rotational barriers of 2−6 kcal/mol have been calculated.[15−17] This rotational barrier may be compared with the value of ca. 3 kcal/mol calculated for rotation about

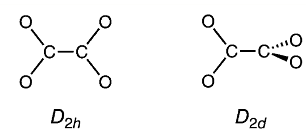

Figure 1. Two structures of the oxalate ion.

the C−C bond in ethane and about the 2,3 C−C bond in *n*-butane.[18] Further, vibrational spectroscopic studies of aqueous solutions of the simple oxalates of the alkali metal cations are consistent with ox^{2-} existing as the D_{2d} conformer in these solutions, for example, refs 19 and 20. A particularly interesting aspect of the theoretical study of Dewar and Zheng[16] was an investigation of the influence of cations (as "disembodied charged spheres") on the stability of the ox^{2-} ion. From this it was found that the dianion changes from the D_{2d} conformer toward the D_{2h} conformer as the size of the cation decreases, reflecting stronger cation−dianion electrostatic interactions and better fit into the "bite" of the ox^{2-} with diminishing cation size. This structural prediction is confirmed by the structures of the anhydrous alkali-metal oxalates.[21] Thus, anhydrous $Cs_2C_2O_4$ contains a twisted ox^{2-} with an O−C−C−O dihedral angle of 81(1)°;[22] anhydrous $Rb_2C_2O_4$ exists in two forms, the α form in which the dihedral angle is 94(3)° and the β form with planar ox^{2-};[22] anhydrous $M_2C_2O_4$ (M = Li, Na, or K) all contain planar ox^{2-}.[22−24] These particular results have been noted in at least one recent textbook,[25] as an example of "intermolecular interactions in the crystal lattice". Interestingly, the concentration dependence of the vibrational spectra of the alkali-metal salts in aqueous solution suggests that the extent of cation−dianion interaction in the solutions increases as cation size decreases, $Cs^+ < Rb^+ < K^+ < Na^+ < Li^+$.[20]

In summary, the ox^{2-} ion is regularly drawn as planar, but there is overwhelming evidence that the bond between the CO_2 moieties is a single bond in the dianion and this allows for rotation of the two planar CO_2 moieties relative to one another. Theoretical studies have shown that for isolated ox^{2-} the D_{2d} conformer is the most stable, and this conformer has been

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shown to persist into aqueous solutions of simple oxalate salts. However, the barrier to C−C rotation is small and may easily be overshadowed by other forces, such as M−O bonding forces in complexes containing bidentate ox^{2-} and, in solid oxalates, "packing forces" due to a variety of intermolecular interactions, for example, electrostatic or hydrogen-bonding interactions, as mentioned above. Therefore in real materials the O−C−C−O dihedral angle of ox^{2-} will commonly be quite different from the 90° of the D_{2d} conformer, as found.

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